LUXSCRIPTURA

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CSI 4130 / 5130: Artificial Intelligence

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Abstract

Currently there is no technologies that are free for analysis of Theology based questions that fixate on Catholicism explicitly. Lux Scriptura provides a solution by being an AI based on Catholicism and analyzing potential issues / concerns regarding it. Through NLP, RAG, and Gemini, Lux Scriptura tackles difficult and long issues by analyzing the question / statement and then connecting it to the Catechism of The Catholic Church. In the end, accuracy is near 100% with some flaws present as the program learns new models.

Problem Statement

Within the Catholic Church and broader Christian discourse, many misunderstandings and misinterpretations arise regarding Catholic teachings. This problem has become more pronounced with the rise of online discussions, social media debates, and generative AI tools that may produce inaccurate theological explanations. This is what ultimately causes the division between Catholics and other groups such as Protestants, Jewish People, etc…

Many individuals—both Catholics and non-Catholics—seek explanations of Church doctrine but struggle to find answers that are both faithful to official teaching and accessible in plain language. While the Catechism of the Catholic Church contains authoritative explanations of doctrine, it is lengthy, structured in theological language, and not always easy for users to interpret or apply to specific questions. LuxScriptura is aimed at resolving these issues alongside many more.

Introduction

I have hands on experiencing defending the Catholic fatih and recgonize what is valued in a conversation verse what is not. Given this information, I know that the Catechism of the Catholic Church and specific answers are required to defend the faith well.

This can be done by implementing LLM API's alongside Modeling. Through NLP and various other methods taught in class, over time the model will develop to correct and inform the user of any possible misunderstandings their statements may promote.

Additionally, to show the effectiveness of LuxScriptura, the correctness of the theology will be graded by percentage. This method is similar to how we code in class with the methods developed independently of provided code is tested for accuracy and effectiveness.

Related Work

Truthly AI

- A premium AI developed to educate Catholics and non-Catholics on Catholicism and Christianity. This is a newer AI that has been briefly released to the market for selected, paid consumers. If a user is to ask a question about Catholicism or say something false of it, it will not only correct you but provide sources.
- The only decide with this application is, the current technology isn't free and requires a large monthly payment to use, which is not practical.

Side Note: *The meaning behind this project steams into my own life as my current job for a marketing / outreach in a Catholic ministry has challenged me to learn more. This application even when developing it has allowed me to grow in my own knowledge of the Catechism as it it very extensive in length.*

Data

Catechism of The Catholic Church (7981 Paragraphs) was used and flattened accordingly to just paragraphs in order to preform properly.

Bible API: King James Version (Catholic Approved Bible Versions Don't Exist ⟶ Unless Manually Made)

Methods

Applications

- **Google Colab**
  - Runtime Type: Python 3
  - Hardware Accelerator: T4 GPU
    - Other GPU's failed or could not fully handle what I was doing.
- **Google GEMNI API**
  - Model: Gemini-2.0-Flash
- **Visual Studio Code**
  - Testing of background code, which was used briefly for coding until I decided to switch over to Google Colab due to its extensive availability of resources / libraries / and CPU functionalities.

Techniques

- **Gemini AI (Model: Gemini-2.0-Flash)**
- **NLP**
- **RAG**
- **Text Classification & Vectors**

Experiments & Results

UPDATING

Conclusion

Citations & Acknowledgements

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